Problem-Solution Fit

Date	03 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Residents in flood-prone regions • Disaster Management Authorities • Emergency Response Teams • Local Government Bodies • Farmers & Agricultural Communities • Environmental Agencies	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** • Limited internet connectivity • Lack of awareness about flood preparedness • Limited access to smart devices • Inadequate real-time monitoring infrastructure • Budget constraints	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **CONS [AS]** **Available Solutions:** Weather forecasting apps, Government alert systems, River monitoring systems. **Pros:** Provide weather information and basic alerts. **Cons:** Limited prediction accuracy, delayed notifications, lack of localized flood risk analysis.

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

- Unexpected floods causing loss of life and property
- Delayed flood warnings
- Inaccurate flood predictions
- Poor disaster preparedness
- Frequently occurs during heavy rainfall and monsoon seasons

9. PROBLEM ROOT / CAUSE [RC]

- Climate change
- Heavy and unpredictable rainfall
- River overflow
- Poor drainage systems
- Deforestation
- Rapid urbanization
- Lack of early warning systems

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- Regularly monitor weather updates during monsoon
- Respond immediately to emergency alerts
- Depend on local authorities for information
- Share warnings with family and community members

3. TRIGGERS TO ACT [TR]

- Heavy rainfall forecasts
- Rapid rise in river water levels
- Severe weather warnings
- Government emergency alerts
- Historical flood trends indicating high risk

4. EMOTIONS
BEFORE / AFTER **[EM]**

Before: Fear, Anxiety, Uncertainty, Stress
After: Confidence, Preparedness, Safety, Peace of Mind

10. YOUR SOLUTION [SL]

Rising Waters is an AI-powered flood prediction system that uses machine learning (XGBoost) to analyze real-time weather conditions, rainfall patterns, river water levels, and historical flood data. It predicts flood risks accurately and sends early warnings and real-time alerts to citizens, emergency response teams, and disaster management authorities, enabling timely evacuation and reducing flood-related losses.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE: Mobile App, Web Portal, SMS, Email, Social Media

OFFLINE: Public Announcements, Radio, Television, Community Volunteers, Local Government Offices